Humanitarian Classification of Disaster Tweets by Fusing Text and Images

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Abstract

The text and image of a disaster tweet often carry different information. We classify CrisisMMD tweets into five humanitarian categories and compare early fusion, late fusion and cross-attention on frozen BERT, EfficientNet-B0 and ResNet-50 features. Fusion raises test macro-F1 from 0.74 for the best single modality to 0.82, and the gain disappears when texts and images are paired at random. The fusion method matters much less: logistic late fusion reaches 0.80, and cross-attention brings no gain. Macro-F1 varies by about 20 points between disaster events, so we recommend simple late fusion with human review.

1 Introduction

During a disaster, tweets can tell relief organisations where people are trapped, what is damaged and who is offering help, but manual screening cannot keep up with the volume, so a classifier should sort them first. The stakeholders are affected people, whose requests may be missed, and relief organisations, which allocate teams by these decisions.

Each sample is one tweet text with one image, to be assigned to one of five categories: affected individuals, infrastructure and utility damage, rescue, volunteering or donation effort, other relevant information, and not humanitarian. Images tend to show damage and text describes rescue and donations, which motivates fusion.

We ask whether fusion improves on the best single modality, which fusion method works best, and whether the result holds for an unseen disaster. The proposal defined success as a fusion model with macro-F1 above 0.80 and at least five points above the best single modality, with a 95% bootstrap interval excluding zero. The code is at `https://github.com/FT1ger/multimodel_final`.

2 Data and Preprocessing

CrisisMMD v2.0 contains 16,058 tweets and 18,082 images from seven disasters in 2017, labelled separately for text and image (Alam et al., 2018). We use the agreed split of Ofli et al. (2020), which keeps the pairs whose two labels match and merges eight classes into five. Table 1 gives its class distribution.

Table 1: Class distribution of the agreed split.

Class	Train	Validation	Test
Affected individuals	71	9	9
Infrastructure and utility damage	612	80	81
Rescue, volunteering or donation effort	912	149	126
Other relevant information	1,279	239	235
Not humanitarian	3,252	521	504
Total	6,126	998	955

The classes are imbalanced (not humanitarian is 53% of the training pairs and affected individuals has nine test samples), so macro-F1 is the main metric and the loss is weighted by inverse class frequency. Text and image labels agree for only 45% of all pairs, so the agreed split drops more than half of the data. We treat 76 placeholder images as missing. Perceptual hashing (Zauner, 2010) and difference hashing (Krawetz, 2013) flag 64 test images with a near copy in the training or validation set. We keep the official split for comparability and check sensitivity (Section 4.5).

Broken character encoding in 784 tweets is repaired with ftfy (Speer, 2019). Retweet prefixes and URLs are removed and user mentions are replaced by a placeholder token. BERT inputs are cut at 96 tokens, enough for tweets averaging 14 words. Images are resized, cropped to the central 224 by 224 pixels and normalised with ImageNet statistics (Deng et al., 2009). Features are cached and standardised with training set statistics.

3 Methods

3.1 Features and fusion models

Figure 1 outlines the pipeline. Text features are the mean of the last BERT layer (Devlin et al., 2019). Image features are the pooled outputs of EfficientNet-B0 (B0) (Tan and Le, 2019) or ResNet-50 (R50) (He et al., 2016). The encoders stay frozen because six thousand training pairs are few for updating them.

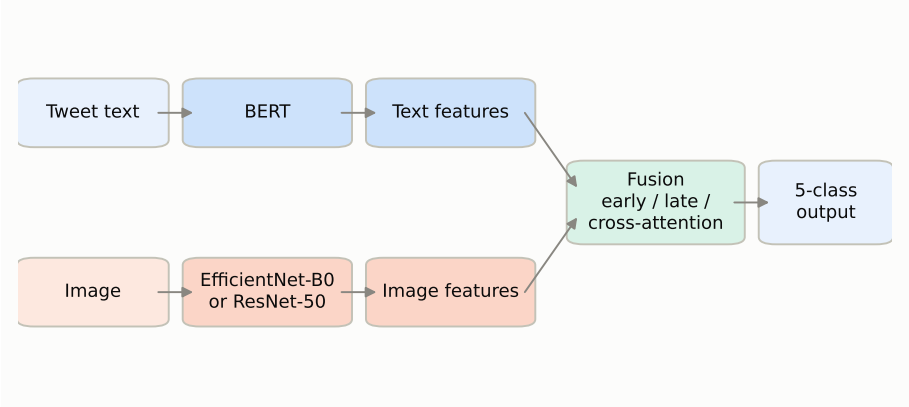


Figure 1: Fusion pipeline.

Following the taxonomy of Baltrušaitis et al. (2019), we compare early fusion, late fusion and cross-attention, where fusion is built into the model. Early fusion concatenates both vectors with a missing image flag and feeds a multilayer perceptron (MLP). It can learn interactions between modalities, but its large input makes it the most likely to overfit. Late fusion trains one classifier

per modality and averages their class probabilities with a learned weight, set to zero for a missing image. It ignores interactions but is robust to a missing modality. Cross-attention (Vaswani et al., 2017) works on tokens, the 49 cells of the B0 feature map and up to 64 BERT token states. One block lets text attend to image tokens and another does the reverse, and a learned token replaces a missing image. It can align words with image regions but has the most parameters.

Logistic regression (LR) on the same features gives simple baselines. Three controls test where a gain comes from: single modality MLPs enlarged to about the size of early fusion, early fusion with texts shuffled across images, and late fusion on averaged tokens. A separate model with two output heads predicts the text and image labels on the full split, which includes disagreeing pairs.

3.2 Training and evaluation

Neural models use cross-entropy weighted by class, AdamW (Loshchilov and Hutter, 2019) with learning rate 10^{-4}, batch size 32, and early stopping on validation macro-F1 with patience seven, over three seeds. Appendix B lists all settings.

We report macro-F1, accuracy, F1 per class and event, a confusion matrix and expected calibration error (ECE) (Guo et al., 2017). Macro-F1 differences come with 95% bootstrap intervals over test rows and seeds (Efron and Tibshirani, 1993), and McNemar's test compares paired accuracy (McNemar, 1947). Overfitting is assessed with learning curves and the gap between training and test macro-F1. In leave-one-event-out (LOEO) evaluation we retrain without one disaster and test on its test pairs. The test set was already scored in the preliminary experiment and thirteen models are compared on it, so test results are indicative.

4 Results

4.1 Fusion against single modalities

Table 2 and Figure 2 show that every fusion model beats every single modality model. The best single model, the R50 image MLP, scores 0.743. Late and early fusion with R50 features reach 0.820 and 0.821, gains of 0.077 (95% interval [0.027, 0.127]) and 0.078 [0.018, 0.133]. Both meet the success criterion, as does B0 early fusion (0.812, gain 0.068 [0.005, 0.129]). LR late fusion misses it at 0.799 with an interval that includes zero. Ofli et al. (2020) report 78.4% accuracy for their multimodal network, against 85% to 88% here.

Table 2: Test results on the agreed split. Gap is training minus test macro-F1; neural rows are means over three seeds. LR gaps come from the rerun described in Appendix C.

Model	Macro-F1	Accuracy	Gap	Parameters
Text only (LR)	0.675	0.757	0.084	4K
Image only (LR, B0)	0.703	0.782	0.090	6K
Early fusion (LR, B0)	0.795	0.863	0.185	10K
Late fusion (LR, B0)	0.799	0.855	0.070	10K
Text only (MLP)	0.726 ± 0.011	0.786	0.255	0.20M
Image only (MLP, B0)	0.711 ± 0.009	0.786	0.162	0.33M
Image only (MLP, R50)	0.743 ± 0.012	0.805	0.170	0.53M
Early fusion (MLP, B0)	0.812 ± 0.016	0.874	0.182	1.18M
Late fusion (MLP, B0)	0.797 ± 0.007	0.847	0.125	0.53M
Late fusion (tokens, B0)	0.798 ± 0.010	0.847	0.126	0.53M
Cross-attention (B0)	0.795 ± 0.019	0.858	0.166	2.27M
Early fusion (MLP, R50)	0.821 ± 0.019	0.880	0.152	1.58M
Late fusion (MLP, R50)	0.820 ± 0.015	0.856	0.104	0.72M

The controls in Table A4 (Appendix C) rule out network size as the cause. An MLP as large as early fusion reaches only 0.721 on images and 0.717 on text, and early fusion with shuffled texts drops from 0.812 to 0.703, the level of images alone.

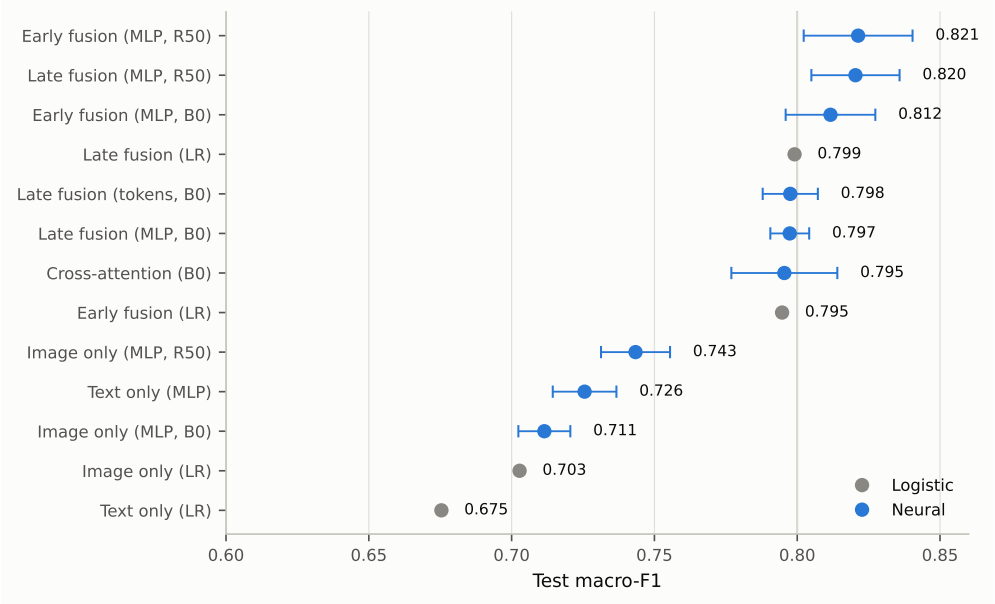


Figure 2: Test macro-F1 by model. Bars show one standard deviation over seeds.

4.2 Comparison of fusion methods

The fusion methods differ little. With B0 features they span 0.795 to 0.812, their pairwise intervals include zero in most seeds, and cross-attention is no better than LR late fusion. Replacing B0 with R50 adds 0.032 for images alone, 0.023 for late fusion and 0.010 for early fusion, small effects that we did not test.

R50 early and late fusion differ by 0.001 in macro-F1, yet early fusion has higher accuracy (0.880 against 0.856, McNemar $p < 0.05$ in every seed). In Figure 3, early fusion is stronger on the large classes and late fusion on the rare affected individuals class (F1 0.73 against 0.62, uncertain with nine samples; Table A5). Fusion raises F1 for infrastructure damage from at most 0.78 to

0.85 or more. In the confusion matrix (Figure A1), 80 of the 138 errors of LR late fusion are not humanitarian tweets predicted as humanitarian; the 37 humanitarian tweets predicted as not humanitarian are the harmful errors. In a rerun of this model, 66 of its 138 errors are pairs that both single modality LR models also miss, and fusion loses no pair that both classify correctly. The 200 predictions with confidence below 0.5 hold 82 of the errors, which gives a simple rule for human review.

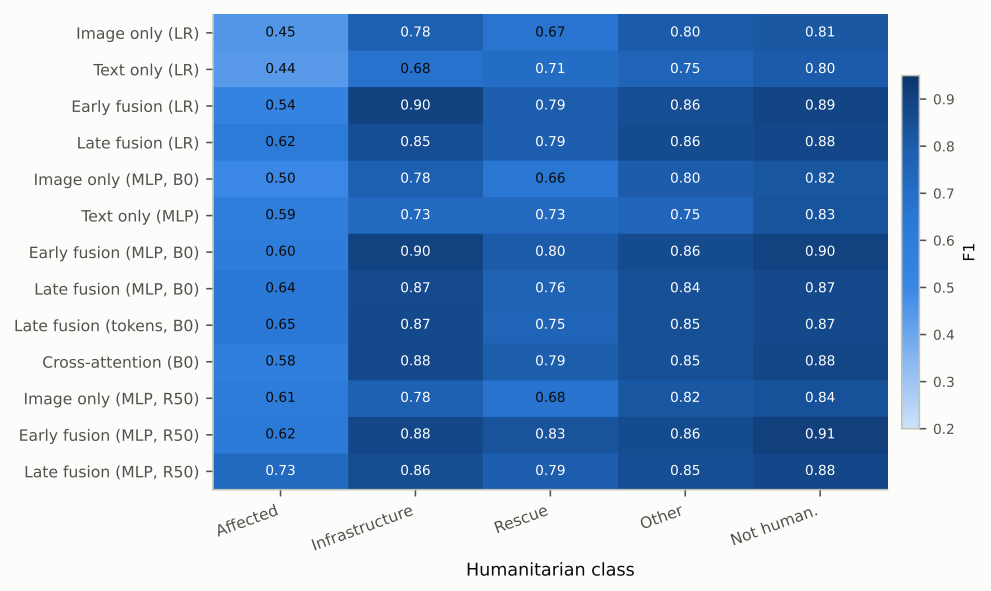


Figure 3: Test F1 per class and model.

4.3 Overfitting

All neural models overfit: training macro-F1 is 0.10 to 0.25 above test macro-F1 (Gap in Table 2). Early fusion has the largest gap among fusion models and late fusion the smallest. In Figure 4 the cross-attention training loss falls from 0.75 to 0.03 while its validation loss rises from 0.53 to above 1.0, and validation macro-F1 of the three fusion methods flattens within ten epochs, so early stopping limits the damage.

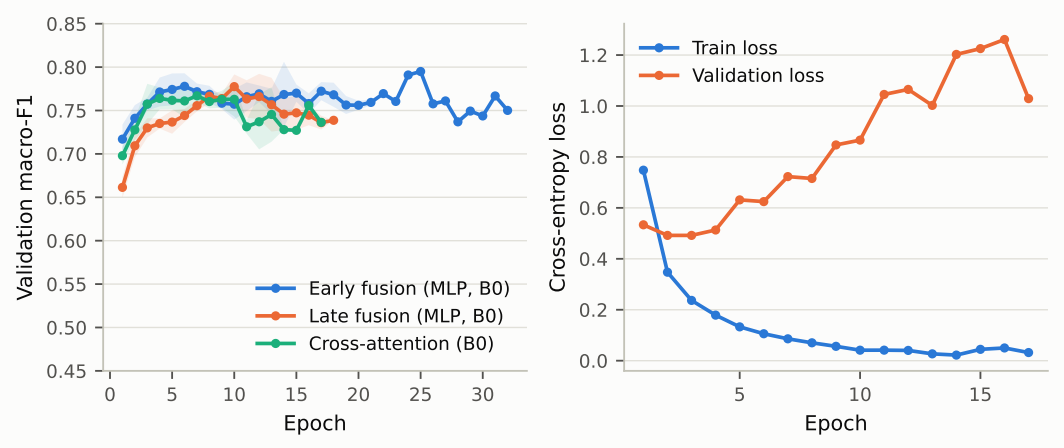


Figure 4: Validation macro-F1 (left) and cross-attention loss for one seed (right).

4.4 Unseen events

Figure 5 gives the LOEO results for the LR models, with macro-F1 computed over the classes present in each event. Fusion still helps on an unseen disaster: mean macro-F1 is 0.764 for early and 0.735 for late fusion against 0.627 for text and 0.599 for images. Early fusion ranges from 0.67 on the Iraq-Iran earthquake to 0.85 on Hurricane Harvey. Relative to the same test pairs with the event in training, the mean change is +0.008 for early and −0.044 for late fusion, and every interval for a single event includes zero (Table A8). The neural models lose 0.024 (B0 early) and 0.057 (R50 late). Leaving an event out therefore costs little; the variation between events already exists on the standard split (Figure A2).

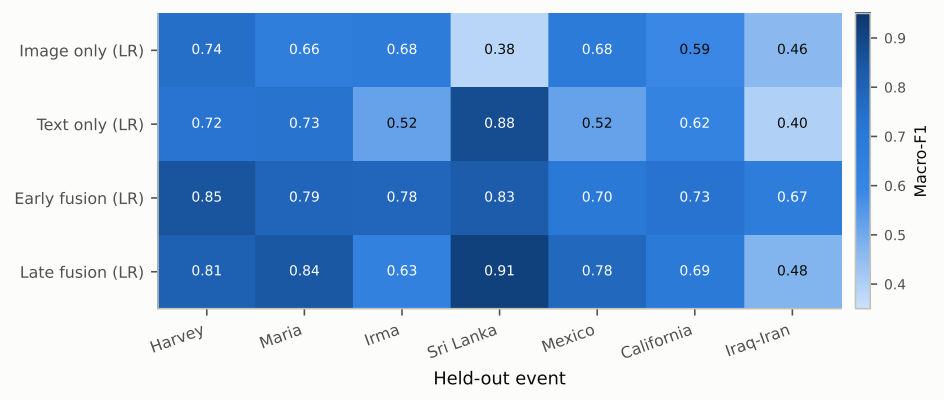


Figure 5: LOEO macro-F1 of the LR models over the classes present in each event.

4.5 Robustness, calibration and cost

Removing the 64 flagged near duplicates changes macro-F1 by at most 0.010, so leakage does not explain the results. The model with two heads reaches 0.62 (text labels) and 0.70 (image labels) on the full test split against 0.67 and 0.75 on its agreed pairs (Table A9). Agreed pairs are therefore easier than real tweet streams. According to Table A6, ECE is 0.185 for LR late fusion and 0.045 to 0.079 for neural fusion, so the LR probabilities need recalibration. Cross-attention takes about eight times longer than neural late fusion to train and to predict.

5 Discussion

Fusing the modalities is the choice that matters, worth about eight points. The image encoder (one to three points) and the fusion method (under two points) have much smaller and uncertain effects. Cross-attention probably cannot use its flexibility with six thousand short tweets and frozen tokens that were never trained to align. Freezing the encoders made the controls and LOEO affordable but lowers the ceiling. Table A1 lists the changes from the proposal; the one that weakens the conclusions is the reuse of the test set.

For a relief organisation we recommend neural late fusion on frozen R50 and BERT features. It matches the best macro-F1, overfits least, handles a missing image by design and is cheap to retrain for a new disaster. It scales to large volumes because feature extraction, the expensive step, runs once per tweet. On a new event, a small labelled sample should be used to check performance.

Limitations are the nine test samples of the rarest class, the reused test set, and seven English language events from one year. Fine-tuning the top encoder layers comes first in future work, since it would show whether cross-attention gains from trainable tokens. Other steps are focal

loss (Lin et al., 2017) for the rare class and training with the disagreeing pairs.

6 Ethics and Bias

Tweet authors did not expect their posts to train a model. The CrisisNLP terms allow research use only, forbid redistribution and require deletion afterwards (CrisisNLP, 2025). We keep the data local, replace usernames and publish only code and aggregate results.

The data is biased towards one kind of disaster. Three Atlantic hurricanes provide 75% of the images and the text is mostly English. In Sri Lanka 87% of pairs are not humanitarian, and the image LR model scores 0.35 there against 0.52 or more on the other events (Figure A2). Four events have 80 or fewer test pairs (Table A8), so the data cannot show whether the model is fair across regions. The labels are a second source of bias, since text and image annotations often differ.

The loss is weighted by class to protect the rare classes. In a control without weights (Table A7), B0 late fusion misses 74 humanitarian tweets instead of 44 and raises 35 false alarms instead of 84, with macro-F1 0.017 higher. We keep the weights because a missed request for help is the most harmful error. With LR late fusion, lowering the threshold for that class to 0.20 finds all nine test cases while flagging 5% of tweets for review. A person should make every final decision. The same classifier could monitor people, so its use should be limited to humanitarian response.

7 Conclusion

Fusing text and images improves humanitarian tweet classification by about eight points of macro-F1 over the best single modality. Early fusion, late fusion and cross-attention perform alike, so late fusion on frozen features is the practical choice. Performance varies by about 20 points between events.

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A Proposal Changes

In the mapping from eight to five classes, injured or dead people and missing or found people join affected individuals, and vehicle damage joins infrastructure and utility damage. Table A1 lists the changes from the proposal with their reasons.

Table A1: Changes from the proposal.

Proposal	Final	Reason and effect
Three image encoders including ViT-B/16 (Dosovitskiy et al., 2021)	EfficientNet-B0 and ResNet-50	Time budget. The encoder comparison has two models.
CLIP (Radford et al., 2021) as a reference model	Not run	Time budget. CLIP is left to future work.
Test set used once	Used in the preliminary experiment and the final evaluation	The preliminary experiment had already scored the logistic baselines on it. Test results are indicative.
Each duplicate group kept in one split	Official split kept, sensitivity check added	Keeps results comparable with Ofli et al. (2020). Removing flagged images changes macro-F1 by at most 0.010.
Two output heads in the main model	Separate small model with two heads	Keeps the main comparison simple. The disagreement question is still answered.
Weighting by event to narrow the gap between events	Not run	Time budget. Scores per event are reported.
About 67 placeholder images	76	A third placeholder type was found after the proposal.
52 near duplicates in the test set, effect under 0.5 points	64, effect up to 1.0 point	Difference hashing was added to perceptual hashing. The conclusion is unchanged.
Attention mask for missing images	Flag, zero weight or learned token	Each fusion method needs its own handling. Only 76 images are affected.

B Training Configuration

Table A2 lists the settings shared by the neural models. The logistic baselines use scikit-learn with balanced class weights.

Table A2: Training configuration.

Setting	Value
Optimiser	AdamW, learning rate 10^{-4}, weight decay 10^{-4}
Batch size	32 (two heads: 64)
Maximum epochs	50 (two heads: 30)
Early stopping patience	7 epochs (two heads: 5)
Dropout	0.2 (two heads and neural LOEO: 0.3)
Hidden size	256 (early fusion: 512 then 256)
Cross-attention	4 heads, feed-forward size 1,024, 49 image and 64 text tokens
Seeds	0, 1, 2
Logistic regularisation C	grid from 10^{-4} to 10
Bootstrap	2,000 resamples
Standardisation	mean and standard deviation of the training set

C Additional Results

Table A3 lists the macro-F1 differences behind Section 4. Intervals for neural models resample seeds and test rows; the logistic rows resample test rows only.

Table A3: Macro-F1 differences with 95% bootstrap intervals.

Comparison	Difference
Late fusion (R50) minus image only (R50)	+0.077 [0.027, 0.127]
Early fusion (R50) minus image only (R50)	+0.078 [0.018, 0.133]
Early fusion (B0) minus image only (R50)	+0.068 [0.005, 0.129]
Late fusion (LR) minus image only (R50)	+0.056 [−0.011, 0.120]
Early fusion (R50) minus late fusion (R50)	+0.001 [−0.034, 0.033]
Late fusion (LR) minus image only (LR)	+0.096 [0.039, 0.151]

Table A4: Controls for network size and pairing (mean over three seeds).

Model	Input	Macro-F1	Parameters
Early fusion (B0)	image and text	0.812	1.18M
Enlarged MLP (B0)	image	0.721	1.18M
Enlarged MLP	text	0.717	1.18M
Early fusion (B0), shuffled texts	image and text	0.703	1.18M
Early fusion (R50)	image and text	0.821	1.58M
Enlarged MLP (R50)	image	0.736	1.77M
Early fusion (R50), shuffled texts	image and text	0.727	1.58M

Table A5: Affected individuals class (nine test samples, mean over seeds).

Model	Recall	Precision	F1
Late fusion (LR)	0.889	0.471	0.615
Early fusion (MLP, B0)	0.704	0.519	0.596
Late fusion (MLP, B0)	0.667	0.622	0.643
Cross-attention (B0)	0.704	0.494	0.579
Early fusion (MLP, R50)	0.778	0.526	0.624
Late fusion (MLP, R50)	0.815	0.661	0.725

In Table A5 late fusion with R50 features has the highest F1, but the bootstrap intervals of its differences from logistic late fusion (+0.110) and from R50 early fusion (+0.101) include zero in every seed.

Table A6: Calibration and cost of the fusion models. Times are relative to late fusion (MLP, B0) and come from one run per model.

Model	ECE	Parameters	Training time	Prediction time
Late fusion (LR)	0.185	10K		
Early fusion (MLP, B0)	0.072	1.18M	0.8	1.1
Late fusion (MLP, B0)	0.079	0.53M	1.0	1.0
Late fusion (tokens, B0)	0.072	0.53M	2.1	1.7
Cross-attention (B0)	0.074	2.27M	8.3	8.5
Early fusion (MLP, R50)	0.045	1.58M	0.9	1.0
Late fusion (MLP, R50)	0.076	0.72M	1.0	1.0

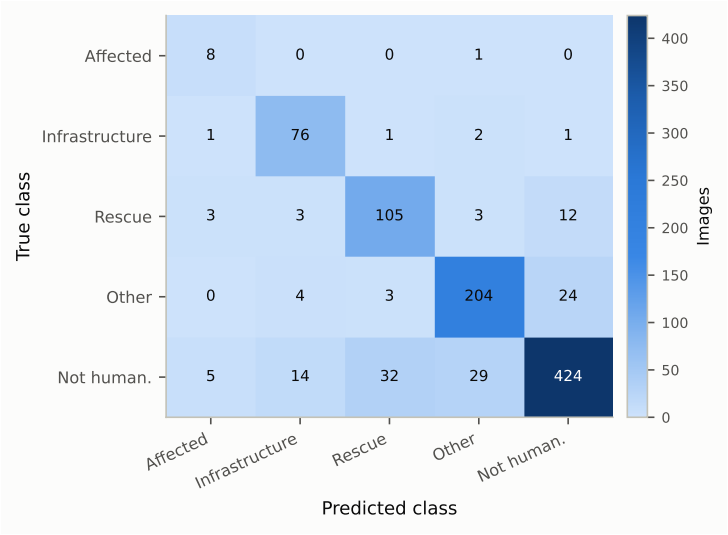


Figure A1: Confusion matrix of logistic late fusion on the test set.

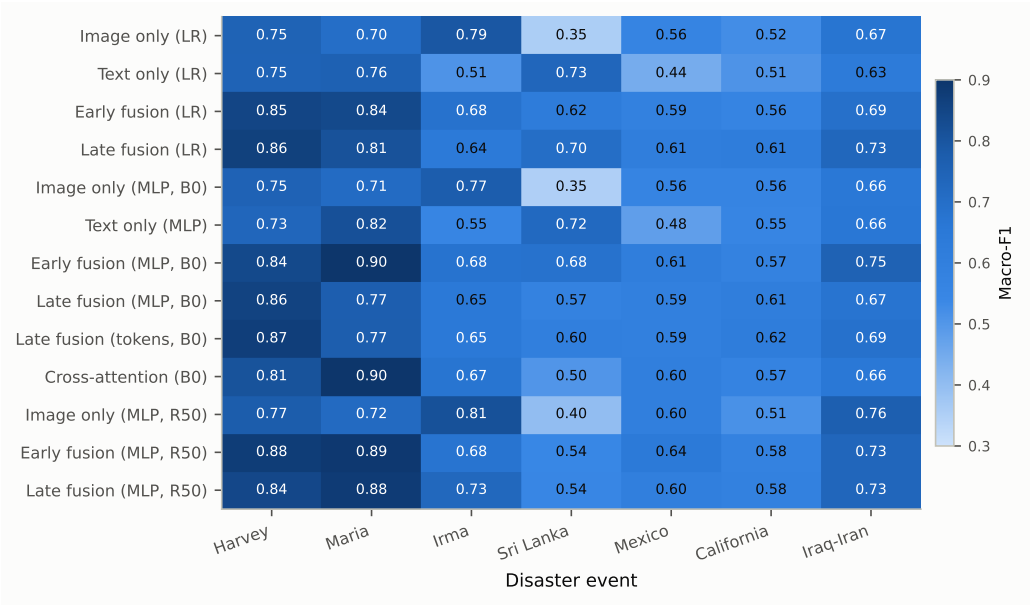


Figure A2: Test macro-F1 per event on the standard split, over all five labels. California, Mexico and Sri Lanka lack one class in the test set, so their scores are capped at 0.80.

Table A7 compares the weighted loss with an unweighted loss. It comes from a separate rerun with newly extracted B0 and BERT features, whose scores differ from Table 2 by less than 0.01, so its rows are compared only with each other. The weights do not raise test macro-F1, but they cut the humanitarian tweets predicted as not humanitarian by a fifth to two fifths at the price of more false alarms. Recall for affected individuals rests on nine samples. The same rerun gives the LR gaps in Table 2 and the error counts in Section 4; its LR late fusion also makes 138 errors, on a slightly different set of pairs.

Table A7: Weighted against unweighted loss with B0 features (mean over three seeds). Missed counts humanitarian tweets predicted as not humanitarian; false alarms are the reverse.

Model	Loss	Macro-F1	Accuracy	Affected recall	Missed	False alarms
Late fusion (MLP)	weighted	0.789 ± 0.004	0.845	0.67	44	84
Late fusion (MLP)	unweighted	0.806 ± 0.011	0.869	0.48	74	35
Early fusion (MLP)	weighted	0.806 ± 0.011	0.876	0.67	43	55
Early fusion (MLP)	unweighted	0.820 ± 0.029	0.878	0.59	55	45

Table A8 gives the LOEO results per event. F1 is macro-F1 over the classes present in the event. The change column compares the LOEO model with the same model trained on the standard split, on the same test pairs and over all five labels.

Table A8: LOEO macro-F1 of logistic early and late fusion, with the change from the standard split and its 95% interval.

Held-out event	n	Early F1	Change	Late F1	Change
California wildfires	78	0.729	+0.019 [−0.028, 0.069]	0.690	−0.063 [−0.148, 0.023]
Hurricane Harvey	248	0.850	+0.001 [−0.045, 0.063]	0.811	−0.052 [−0.239, 0.079]
Hurricane Irma	215	0.782	+0.098 [−0.030, 0.210]	0.635	−0.008 [−0.043, 0.026]
Hurricane Maria	232	0.788	−0.050 [−0.121, 0.005]	0.840	+0.029 [−0.039, 0.110]
Iraq-Iran earthquake	23	0.666	−0.027 [−0.173, 0.105]	0.476	−0.253 [−0.425, 0.005]
Mexico earthquake	79	0.699	−0.028 [−0.100, 0.037]	0.777	+0.010 [−0.030, 0.053]
Sri Lanka floods	80	0.830	+0.040 [−0.069, 0.210]	0.913	+0.029 [−0.096, 0.154]
Mean		0.764	+0.008	0.735	−0.044

Table A9: F1 per class of the model with two output heads on the full test split (mean over three seeds).

Class	Text n	Text F1	Image n	Image F1
Affected individuals	143	0.563	101	0.555
Infrastructure and utility damage	171	0.511	494	0.807
Rescue, volunteering or donation effort	454	0.713	256	0.602
Other relevant information	844	0.659	331	0.711
Not humanitarian	625	0.678	1,055	0.808
Macro-F1, full test split		0.625		0.697
Macro-F1, agreed test pairs		0.670		0.745

According to Table A9, the text head is strongest on rescue and the image head on infrastructure damage, which matches the complementary roles of the two modalities in the fused models. On the 1,267 test pairs whose labels disagree, the text head matches the text label for 57% and the image head matches the image label for 70%.